

exponential moving average:

$$\left(P_t \frac{2}{N+1}\right) + \left(ema_{t-1} \left(1 - \frac{2}{N+1}\right)\right)$$

simple moving average:

$$\frac{1}{N} \sum_{i=0}^{N-1} P_{t-i}$$

bollinger bands:

$$\frac{1}{N} \sum_{i=0}^{N-1} P_{t-i} \pm \left(k \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} \left(P_{t-i} - \frac{1}{N} \sum_{i=0}^{N-1} P_{t-i} \right)^2} \right)$$

relative strength index:

$$rsi_t = 100 - \left(\frac{100}{1 + \frac{(\text{profit}_n \frac{2}{N+1}) + (ema_{t-1} (1 - \frac{2}{N+1}))}{(\text{loss}_n \frac{2}{N+1}) + (ema_{t-1} (1 - \frac{2}{N+1}))}} \right)$$

volume ratio:

$$\frac{volume_t}{\frac{1}{N} \sum_{i=0}^{N-1} volume_{t-i}}$$